Self-Triggered Nash Equilibrium Seeking for Multi-Player Linear Quadratic Differential Games

Tianhao Fei, Yongliang Yang*
School of Automation and Electrical Engineering
University of Science and Technology Beijing
Beijing, China
feith@ieee.org, yangyongliang@ieee.org

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***Abstract*—This paper studies continuous-time linear systems with multiple control inputs by formulating them as a nonzero-sum game. The Nash equilibrium is characterized via a set of coupled algebraic Riccati equations, and an offline policy iteration is implemented to compute the feedback gains. However, directly implementing the Nash feedback requires continuous state measurement, which can be prohibitive in resource-limited systems. To overcome this intrinsic limitation of event-triggered control that relies on continuous state monitoring to check the triggering condition, a novel self-triggered control mechanism is proposed. By relating the time-varying state and sampling error to the sampled state, the next triggering instant can be computed explicitly without continuous observation, thereby further reducing communication overhead. Theoretical analysis and simulation results confirm the efficiency of the proposed self-triggered control method for the nonzero-sum game.**

***Index Terms*—Nash equilibrium, nonzero-sum game, self-triggered control, coupled algebraic Riccati equations.**

I. Introduction

Nonzero-sum (NZS) differential games provide a principled framework to model multi-agent decision-making in dynamical systems, where each player optimizes its own performance while interacting with others [1], [2]. For continuous-time linear systems with quadratic criteria, the Nash equilibrium can be characterized by a set of coupled Hamilton–Jacobi (cHJ) equations [3], [4] and further reduced to coupled algebraic Riccati equations (cAREs) by exploiting the quadratic value-function structure [5]. To compute the coupled Riccati solutions and the associated equilibrium feedback gains, one may employ policy-iteration schemes widely used in adaptive dynamic programming (ADP) [6]–[8], which alternate between policy evaluation and policy improvement until convergence. This iterative structure is closely related to the Newton–Kleinman method for Riccati-type iterations [9], [10].

Despite the appeal of continuous-state Nash feedback, its implementation typically presumes continuous sensing and communication, which is often unrealistic in networked or resource-constrained settings where bandwidth, energy, and computation are inherently limited. Event-triggered control (ETC) [11], [12] addresses this issue by updating the control input only when a state-dependent triggering condition is violated, rather than enforcing periodic sampling regardless of whether the system state has meaningfully changed. This mechanism reduces unnecessary sampling and transmissions, alleviates network congestion, and can significantly lower the sensing/communication load while still preserving closed-loop stability and excluding Zeno behavior via a strictly positive inter-event time [13], [14]. However, ETC still has an intrinsic limitation: the triggering rule must be checked online, which requires continuous state monitoring to decide whether the condition is met, and this can remain a significant communication burden in resource-limited environments [15].

Self-triggered control (STC) is introduced to remove this continuous-monitoring requirement by predicting the next update instant at the current triggering time [16], [17]. In particular, by establishing explicit relations that bound the time-varying state and sampling error using the time-invariant sampled state over each inter-event interval, the triggering condition can be reformulated so that the next triggering instant is obtained by solving an equation computed at the last event time, thereby eliminating the reliance on continuous state observation [15], [18], [19]. Building on this idea, this paper focuses on integrating a self-triggered mechanism with NZS game-based Nash feedback design: after computing the equilibrium gains from the cAREs, the proposed STC strategy explicitly specifies trigger-time computation and state propagation/update without continuous sensing.

The primary contributions of this paper are twofold.

1) Compared with existing ETC strategies [13], [14] that need continuous state monitoring, this paper proposes a novel STC mechanism. By predetermining the triggering instants, the STC eliminates the requirement for continuous state measurement, thereby reducing computational burden for resource-limited systems.
2) Most existing STC studies are restricted to single-input settings [18], [19]. This paper extends the STC framework to multi-player NZS differential games, where each player's cost functional is minimized under the STC-based implementation of the Nash feedback policies.

The remainder of this paper is arranged as follows. Section II states the NZS game problem. The solution to the NZS game are investigated in Section III. The main result of this paper lies in Section IV. Simulation studies are presented in Section V, followed by the conclusions in Section VI.

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II. Problem Statement

Consider the continuous-time linear dynamical system with N inputs given by:

$$\dot{x}(t) = Ax(t) + \textstyle\sum_{i=1}^{N} B_i u_i(t), \quad x(0) = x_0, \tag{1}$$

where $x(t) \in \mathbb{R}^n$ is the state, each $u_i(t) \in \mathbb{R}^{m_i}$ is the input signal, $A \in \mathbb{R}^{n\times n}$ and $B_i \in \mathbb{R}^{n\times m_i}$ are the plant and input matrices such that the pairs (A, B_i) are stabilizable [9].

In this paper, we view each control input u_i as a *player*, and the other $N-1$ players are denoted as $u_{-i} \triangleq \{u_j | j \in \mathcal{N}, j \neq i\}$, where $\mathcal{N} \triangleq \{1, 2, \cdots, N\}$ is the set of player indices. Each player aims to minimize the cost functional:

$$\mathcal{J}_i(u_i, u_{-i}; x_0) = \int_0^{\infty} r_i\big(x(t), u_i(t), u_{-i}(t)\big)\mathrm{d}t, \tag{2}$$

where $r_i(x, u_i, u_{-i})$ is the instantaneous cost given by

$$r_i(x, u_i, u_{-i}) \triangleq x^\top Q_i x + \textstyle\sum_{j=1}^{N} u_j^\top R_{ij} u_j \tag{3}$$

with $Q_i \succ 0$ and $R_{ij} \succ 0$ for all $i, j \in \mathcal{N}$. The following standard assumption is adopted throughout this paper.

Assumption 1. *[5] The pair $(\sqrt{Q_i}, A)$ is detectable, $\forall i \in \mathcal{N}$.*

According to the dynamic game theory [1], [2], the NZS game associated with (1) and (2) can be formulated as

$$\min_{u_i} \mathcal{J}_i(u_i, u_{-i}; x_0), \ \forall i \in \mathcal{N}. \tag{4}$$

For all u_i, the *Nash equilibrium* $\{u_i^\star, u_{-i}^\star\}$ of (4) satisfies

$$\mathcal{J}_i(u_i^\star, u_{-i}^\star; x_0) \leq \mathcal{J}_i(u_i, u_{-i}^\star; x_0), \ \forall i \in \mathcal{N}, \forall x_0 \in \mathbb{R}^n. \tag{5}$$

The outcome of the NZS game is denoted as $\{\mathcal{V}_i^\star\}_{i=1}^N$:

$$\mathcal{V}_i^\star(x) = \min_{u_i} \int_t^{\infty} r_i\big(x(\tau), u_i(\tau), u_{-i}(\tau)\big)\mathrm{d}\tau, \ \forall x \in \mathbb{R}^n \tag{6}$$

which is positive definite and solves the cHJ equation [4], [5]:

$$\begin{aligned} 0 = \nabla^\top \mathcal{V}_i^\star \Big(Ax - \frac{1}{2}\textstyle\sum_{j=1}^{N} B_j R_{jj}^{-1} B_j^\top \nabla \mathcal{V}_i^\star\Big) + x^\top Q_i x \\ + \frac{1}{4}\textstyle\sum_{j=1}^{N} \nabla^\top \mathcal{V}_j^\star B_j R_{jj}^{-1} R_{ij} R_{jj}^{-1} B_j^\top \nabla \mathcal{V}_j^\star, \quad \forall x \in \mathbb{R}^n. \end{aligned} \tag{7}$$

Since the system (1) is linear, $\mathcal{V}_i^\star(x)$ is in a quadratic form:

$$\mathcal{V}_i^\star(x) = x^\top P_i^\star x, \quad \forall i \in \mathcal{N}. \tag{8}$$

where $P_i^\star = (P_i^\star)^\top \succ 0$. Inserting (8) into (7), the cHJ equations can be reduced to the following cARE [5]:

$$0 = \bar{A}^\top P_i^\star + P_i^\star \bar{A} + Q_i + \textstyle\sum_{j=1}^{N} P_j^\star B_j R_{jj}^{-1} R_{ij} R_{jj}^{-1} B_j^\top P_j^\star, \tag{9}$$

where $\bar{A} \triangleq A - \sum_{j=1}^{N} B_j R_{jj}^{-1} B_j^\top P_j^\star$. Then, the cARE (9) can be solved for $\{P_i^\star\}_{i=1}^N$, and the optimal feedback control for each player can be given by

$$u_i^\star(x) = K_i^\star x, \quad \forall i \in \mathcal{N} \tag{10}$$

with the feedback gain $K_i^\star \triangleq -R_{ii}^{-1} B_i^\top P_i^\star$.

III. Solution to the cAREs

The approximate solution to the set of cAREs (9) can be obtained by *policy iteration* consisting of two steps, namely policy evaluation and policy improvement, as summarized in the following pseudo-code algorithm.

Algorithm 1 Offline Policy Iteration for solving the cAREs

Initialization:
Set iteration step $k \leftarrow 0$;
Set a small $\epsilon \in \mathbb{R}^+$ as the termination tolerance;
Start with a stabilizing tuple $\{K_i^0\}_{i=1}^N$ for $u_i^0 = K_i^0$.
while $\max_i \big\{\|P_i^{k+1} - P_i^k\|\big\} \geq \epsilon$ **do**
  for each i from 1 to N **do**
    Policy Evaluation: Solve the N-uncoupled Lyapunov equations for $\{P_i^k\}_{i=1}^N$ with $\{K_i^k\}_{i=1}^N$:

$$\begin{aligned} 0 =& \big(A + \textstyle\sum_{j=1}^{N} B_j K_j^k\big)^\top P_i^k + P_i^k \big(A + \textstyle\sum_{j=1}^{N} B_j K_j^k\big) \\ &+ Q_i + \textstyle\sum_{j=1}^{N} K_j^\top R_{ij} K_j. \end{aligned} \tag{11}$$

    Policy Improvement: Update the feedback gain:

$$K_i^{k+1} \leftarrow -R_{ii}^{-1} B_i^\top P_i^k. \tag{12}$$

  end for
  Update iteration step: $k \leftarrow k+1$.
end while

The iterative method in Algorithm 1 is related to the quasi-Newton iteration [4], [20]. Specifically, for a given stabilizing gain set $\{K_i^k\}_{i=1}^N$ policy evaluation solves N Lyapunov equations, yielding $\{P_i^k\}_{i=1}^N$ that are well-defined and positive definite under the stabilizing initialization [1], [5]. Policy improvement then updates each feedback gain via (12) which can be interpreted as an approximate Newton-type step [10].

With a sufficiently small ϵ, after some iterations k, we assume the solution $\{P_i^\star\}_{i=1}^N$ and thus $\{K_i^\star\}_{i=1}^N$ can be obtained by Algorithm 1. It should be emphasized, however, that even with the desired gains $\{K_i^\star\}_{i=1}^N$, the Nash equilibrium $\{u_i^\star\}_{i=1}^N$ in (10) requires continuous measurement of the state $x(t)$. However, this can bring communication burden in resource-limited systems [13], [14], [18].

To address this practical limitation while retaining the stability guarantees implied by the offline design, we next develop two triggering mechanisms for the dynamical system (1) with the NZS game (4).

IV. Main Result

A. Sampling Process

Before introducing the triggering strategies, we first denote an increasing time sequence $\langle t_h \rangle$, $h \in \mathbb{N}$ as the execution instants. Accordingly, the sampled state (i.e., the intermittent feedback for the players) is defined as follows.

$$\hat{x}(t) = \begin{cases} x(t_h) & , t \in [t_h, t_{h+1}) \\ x(t_{h+1}), & t = t_{h+1} \end{cases} \tag{13}$$

with the sample error given by $\tilde{x}(t) = x(t) - \hat{x}(t)$ and the sample interval $\Delta t_h \triangleq t_{h+1} - t_h$.

Under the sample process (13), the players $\{u_i^\star\}_{i=1}^N$ determined by (10) are implemented with intermittent feedback:

$$u_i^\star(\hat{x}) = K_i^\star \hat{x}, \quad \forall i \in \mathcal{N}. \tag{14}$$

The triggering mechanisms are inspired by the input-to-state stability analysis [11], [12], [16], [21]. To this end, we choose the Lyapunov candidate of the system (1) as

$$\mathcal{V}(x) \triangleq \textstyle\sum_{i=1}^N \mathcal{V}_i(x) = x^\top P^\star x, \quad P^\star \triangleq \sum_{i=1}^N P_i^\star. \tag{15}$$

Then, $\mathcal{V}(x)$ is radially unbounded [21] and satisfies

$$\lambda_{\min}(P^\star)\|x\|^2 \leq \mathcal{V}(x) \leq \lambda_{\max}(P^\star)\|x\|^2, \quad \forall x \in \mathbb{R}^n \tag{16}$$

Differentiating $\mathcal{V}_i(x)$ along the dynamics (1) yields

$$\begin{aligned}\dot{\mathcal{V}}_i(x) =& 2x^\top P_i^\star \big[Ax + \textstyle\sum_{j=1}^N B_j \hat{u}_j^\star(\hat{x})\big] \\ =& 2x^\top P_i^\star \textstyle\sum_{j=1}^N B_j \big[u_j^\star(\hat{x}) - u_j^\star(x)\big] \\ &+ 2x^\top P_i^\star \big[Ax + \textstyle\sum_{j=1}^N B_j \hat{u}_j^\star(x)\big]\end{aligned} \tag{17}$$

Substituting (10) and (14) into (17) gives

$$\dot{\mathcal{V}}_i(x) = 2x^\top M_i \tilde{x} + x^\top (\bar{A}^\top P_i^\star + P_i^\star \bar{A})x, \tag{18}$$

where $M_i \triangleq P_i^\star \sum_{j=1}^N B_j R_{jj}^{-1} B_j^\top P_j^\star \succeq 0$. By the cARE (9), one has the following equation

$$\bar{A}^\top P_i^\star + P_i^\star \bar{A} = -Q_i - \textstyle\sum_{j=1}^N P_j^\star B_j R_{jj}^{-1} R_{ij} R_{jj}^{-1} B_j^\top P_j^\star.$$

Then, (17) can be rewritten as

$$\dot{\mathcal{V}}_i(x) = 2x^\top M_i \tilde{x} - x^\top \big(Q_i + \textstyle\sum_{j=1}^N P_j^\star B_j R_{jj}^{-1} R_{ij} R_{jj}^{-1} B_j^\top P_j^\star\big)x,$$

which can be relaxed to following inequality

$$\begin{aligned}\dot{\mathcal{V}}_i(x) &\leq -\lambda_{\min}(Q_i)\|x\|^2 + 2\lambda_{\max}(M_i)\|x\|\|\tilde{x}\| \\ &\leq -\Big[\lambda_{\min}(Q_i) - \frac{\lambda_{\max}(M_i)}{\mu_i}\Big]\|x\|^2 + \mu_i \lambda_{\max}(M_i)\|\tilde{x}\|^2,\end{aligned} \tag{19}$$

where the second inequality comes from the Young's inequality with $\mu_i > 0$.

In what follows, two triggering mechanisms will be established from (19).

B. Event-triggered Mechanism

In this subsection, we investigate the event-driven scheme for the system (1) and the NZS game (4). Based on (19), the triggering condition is designed as

$$\|\tilde{x}(t)\| \leq \sqrt{\frac{\alpha \sum_{i=1}^N \lambda_{\min}(Q_i)}{\sum_{i=1}^N \mu_i \lambda_{\max}(M_i)}} \cdot \|x(t)\| \tag{20}$$

with $\alpha \in (0, 1)$. The event-based sample process satisfies the following standard assumption.

Assumption 2. *[11], [12] Finite-time stabilization is not obtained at sample instants t_h, i.e., $\|\hat{x}(t_h)\| \neq 0$, $\forall h \in \mathcal{N}$.*

Then, under the event-triggering condition (20), the system performance can be investigated in the following theorem.

Theorem 1. *Let the sample rule be* (13) *and the event execution condition be* (20)*. Then, under Assumptions 1 and 2, with the intermittent feedback policies in* (14)*:*

1) *the state $x(t)$ of the system* (1) *converges to the origin exponentially;*
2) *the execution interval Δt_h for all $h \in \mathbb{N}$ has a positive lower bound, i.e., the Zeno phenomenon is excluded.*

Proof. We prove the theorem step by step.

(1) First, recall the Lyapunov function in (15) and (19). The derivative of $\mathcal{V}(x)$ along the dynamics (1) satisfies:

$$\begin{aligned}\dot{\mathcal{V}}(x) \leq& -\Big(\bar{\alpha} \textstyle\sum_{i=1}^N \lambda_{\min}(Q_i) - \sum_{i=1}^N \dfrac{\lambda_{\max}(M_i)}{\mu_i}\Big)\|x\|^2 \\ &- \alpha \textstyle\sum_{i=1}^N \lambda_{\min}(Q_i)\|x\|^2 + \sum_{i=1}^N \mu_i \lambda_{\max}(M_i)\|\tilde{x}\|^2,\end{aligned}$$

where $\bar{\alpha} = 1 - \alpha \in (0, 1)$. Substituting the triggering condition (20) into above inequality, the following holds for all $x \in \mathbb{R}^n$:

$$\dot{\mathcal{V}}(x) \leq -\kappa \|x\|^2, \tag{21}$$

where $\kappa \triangleq \bar{\alpha} \sum_{i=1}^N \lambda_{\min}(Q_i) - \sum_{i=1}^N \frac{\lambda_{\max}(M_i)}{\mu_i}$ can be made positive by selecting proper Q_i, R_{ij}, α and μ_i.

Based on (15) and (21), one has

$$\|x(t)\| \leq \|x_0\| \sqrt{\frac{\lambda_{\max}(P^\star)}{\lambda_{\min}(P^\star)}} \cdot \mathrm{e}^{-\frac{\kappa t}{2\lambda_{\max}(P^\star)}}, \quad \forall t \geq 0, \tag{22}$$

which indicates $x(t)$ converges to the origin exponentially such that the first statement holds.

(2) Next, we analyze the Zeno-free property of the event-triggered mechanism. From Assumption 2 and the condition (20), when the event is triggered, i.e., $t = t_h^-$, one has

$$\frac{\|\tilde{x}(t)\|}{\|x(t)\|} = \sqrt{\frac{\alpha \sum_{i=1}^N \lambda_{\min}(Q_i)}{\sum_{i=1}^N \mu_i \lambda_{\max}(M_i)}} \triangleq \eta. \tag{23}$$

This implies $\frac{\|\tilde{x}(t)\|}{\|x(t)\|}$ increases from 0 to η during $[t_h, t_{h+1})$ and reset to 0 at $t = t_{h+1}$. By [11, Theorem III.1], the dynamics of $\frac{\|\tilde{x}(t)\|}{\|x(t)\|}$ satisfies

$$\frac{\mathrm{d}}{\mathrm{d}t} \frac{\|\tilde{x}(t)\|}{\|x(t)\|} \leq \big(\|A\| + \textstyle\sum_{i=1}^N \|B_i\|\|K_i^\star\|\big)\Big(1 + \dfrac{\|\tilde{x}(t)\|}{\|x(t)\|}\Big)^2 \tag{24}$$

On the interval $[t_h, t)$, applying the comparison lemma [21, Lemma 3.4] to (24) yields

$$\frac{\|\tilde{x}(t)\|}{\|x(t)\|} \leq \frac{(t - t_h)\big(\|A\| + \sum_{i=1}^N \|B_i\|\|K_i^\star\|\big)}{1 - (t - t_h)\big(\|A\| + \sum_{i=1}^N \|B_i\|\|K_i^\star\|\big)}. \tag{25}$$

For $t = t_{h+1}$, (25) implies a positive lower bound of Δt_h:

$$\Delta t_h = t_{h+1} - t_h \geq \frac{\eta}{\big(\|A\| + \sum_{i=1}^N \|B_i\|\|K_i^\star\|\big)(1 + \eta)} > 0.$$

which excludes the Zeno behavior. □

Notice that although the ETC strategy guarantees system stability and Zeno-free property, the triggering condition (20) requires continuous state monitoring to evaluate if the condition (20) is met. This can still increase communication burden. To completely overcome this difficulty, we propose a novel STC mechanism, as detailed in the following subsection.

C. Self-triggered Mechanism

Before proceeding to the design of the self-triggered mechanism, we first make a standard assumption.

Assumption 3. *[16], [17] For all $x \in \mathbb{R}^n$ with its sample $\hat{x}$ and sampling error $\tilde{x}$, there exists $\varrho > 0$, such that*

$$x^\top \big[Ax + \textstyle\sum_{i=1}^N B_i u_i^\star(\hat{x})\big] \leq \varrho\big(1 + \|x\|^2 + \|\tilde{x}\|^2\big),$$
$$\tilde{x}^\top \big[Ax + \textstyle\sum_{i=1}^N B_i u_i^\star(\hat{x})\big] \leq \varrho\big(1 + \|x\|^2 + \|\tilde{x}\|^2\big).$$

This leads to the following lemma.

Lemma 1. *[16], [17] Under Assumption 3, the following inequalities hold*

$$\|\tilde{x}(t)\|^2 \leq \psi_1(\|\hat{x}\|, t - t_h), \tag{26}$$
$$\|x(t)\|^2 \geq \psi_2(\|\hat{x}\|, t - t_h), \tag{27}$$

where

$$\psi_1(\|\hat{x}\|, t - t_h) \triangleq \frac{2\|\hat{x}\|^2 + 1}{3}(\mathrm{e}^{6\varrho t} - 1),$$
$$\psi_2(\|\hat{x}\|, t - t_h) \triangleq \frac{5\|\hat{x}\|^2 + 1}{3}\mathrm{e}^{-6\varrho t} - \frac{2\|\hat{x}\|^2 + 1}{3}.$$

Proof. The proof can be found in [16, Lemma 2]. □

The above lemma establishes the relation of time-varying signals $\tilde{x}(t)$ and $x(t)$ with the time-invariant sampling $\hat{x}(t)$ over the interval $[t_h, t_{h+1})$ (e.g., see (13)). This can be applied to the condition (20) to eliminate the reliance on continuous state observation and thus relax communication overhead.

Based on the discussions set forth, we formulate the self-triggered mechanism as

$$\mathcal{S}_2\big(\psi_1(\|\hat{x}\|, t - t_h)\big) \leq \sigma_1 \mathcal{S}_1\big(\psi_2(\|\hat{x}\|, t - t_h) + \sigma_2\big) \tag{28}$$

with user-defined positive parameters σ_1, σ_2 and the functions $\mathcal{S}_1 \in \mathcal{K}$, $\mathcal{S}_2 \in \mathcal{K}_\infty$ satisfying $\mathcal{S}_1(\|x\|) \leq \|x\| \leq \mathcal{S}_2(\|x\|)$ for all $x \in \mathbb{R}^n$. Therefore, once σ_1, σ_2, $\mathcal{S}_1$, and $\mathcal{S}_2$ are determined, one can directly solve the triggering instant next to t_h from

$$\mathcal{S}_2\big(\psi_1(\|\hat{x}\|, t - t_h)\big) = \sigma_1 \mathcal{S}_1\big(\psi_2(\|\hat{x}\|, t - t_h) + \sigma_2\big). \tag{29}$$

By the class-$\mathcal{K}$ functions $\mathcal{S}_1$ and $\mathcal{S}_2$, and the inequalities in Lemma 1, the self-triggering condition (28) guarantees the following inequality

$$\begin{aligned} \|\tilde{x}\|^2 &\leq \mathcal{S}_2(\|\tilde{x}\|^2) \leq \mathcal{S}_2\big(\psi_1(\|\hat{x}\|, t - t_h)\big) \\ &\leq \sigma_1 \mathcal{S}_1\big(\psi_2(\|\hat{x}\|, t - t_h) + \sigma_2\big) \\ &\leq \sigma_1 \mathcal{S}_1\big(\|x\|^2 + \sigma_2\big) \leq \sigma_1\|x\|^2 + \sigma_1\sigma_2. \end{aligned} \tag{30}$$

Collecting above discussions, we are ready to state the primary results of this paper. The self-triggered mechanism is illustrated in Algorithm 2, which explicitly specifies the initialization, trigger-time computation, and state propagation/update steps. In particular, at each triggering instant t_h, the next update time t_{h+1} is determined by solving (29), and the system state and players are subsequently updated over $[t_h, t_{h+1})$.

Algorithm 2 Self-triggered mechanism for the NZS game

```
Initialization:
Select parameters ϱ, σ_1 and σ_2 in the condition (28);
Set h ← 0, initial time t_h ← 0 and run time t_span;
Set the initial states x(t_h) randomly;
Set x̂(t_h) ← x(t_h);
Start with {K_i^⋆}_{i=1}^N obtained by Algorithm 1 for u_i^⋆(x̂).
while t < t_span do
    Compute the trigger instant t_{h+1} by (29);
    Propagate t and x(t) with (1) and (14) by the ordinary
    differential equation solver on the interval [t_h, t_{h+1}];
    Update the sampled state: x̂(t) ← x(t_{h+1});
    Update the players: u_i^⋆(x̂) ← K_i^⋆ x̂ for all i ∈ 𝒩;
    Update the trigger instant: t_h ← t_{h+1} (or h ← h + 1).
end while
```

The following theorem establishes the stability of the system (1) and guarantees the Zeno-free property of the self-triggering condition (28) under Algorithm 2.

Theorem 2. *Let Assumptions 1 and 3 hold. Consider the players given by* (14) *with intermittent feedback determined by the sample process* (13) *and the self-triggering mechanism* (28). *Then, the following statements can be established:*

1) *the state $x(t)$ of the system* (1) *is uniformly ultimately bounded (UUB);*
2) *the Zeno phenomenon is excluded.*

Proof. We prove the theorem step by step.

(1) First, we show the state is UUB. Inserting (30) into the upper-bounded $\dot{\mathcal{V}}_i(x)$ in (19) yields

$$\begin{aligned} \dot{\mathcal{V}}_i(x) \leq &- \lambda_{\min}(Q_i)\|x\|^2 + \frac{\lambda_{\max}(M_i)}{\mu_i}\|x\|^2 \\ &+ \mu_i\sigma_1\lambda_{\max}(M_i)\|x\|^2 + \mu_i\sigma_1\sigma_2\lambda_{\max}(M_i). \end{aligned} \tag{31}$$

This implies $\dot{\mathcal{V}}(x)$ satisfies

$$\dot{\mathcal{V}}(x) \leq -\gamma_1\|x\|^2 - \gamma_2\|x\|^2 + \gamma_3, \tag{32}$$

where

$$\gamma_1 \triangleq \textstyle\sum_{i=1}^N \Big[\bar{\alpha}\lambda_{\min}(Q_i) - \dfrac{\lambda_{\max}(M_i)}{\mu_i} - \sigma_1\mu_i\lambda_{\max}(M_i)\Big],$$
$$\gamma_2 \triangleq \alpha\textstyle\sum_{i=1}^N \lambda_{\min}(Q_i), \quad \gamma_3 \triangleq \sigma_1\sigma_2\sum_{i=1}^N \mu_i\lambda_{\max}(M_i).$$

Thus, $\dot{\mathcal{V}}(x) \leq 0$, if $\|x\| \geq \sqrt{\gamma_3/\gamma_2}$.

By [21, Theorem 4.18] and the radial unboundedness of $\mathcal{V}(x)$ in (16), the state $x(t)$ is UUB such that

$$\|x(t)\| \leq \frac{\lambda_{\max}(P^\star)}{\lambda_{\min}(P^\star)}\sqrt{\frac{\gamma_3}{\gamma_2}}, \quad \forall t \geq T_x \tag{33}$$

with some time $T_x > 0$.

(2) It should be noted that the triggering instant t_{h+1} can be uniquely determined by (29), since ψ_1 and ψ_2 are monotonically increasing and decreasing in t. That is, the next triggering instant t_{h+1} is uniquely determined.

Next, we show that $\Delta t_h > 0$. Consider a function $\beta(\cdot)$ such that $0 < \beta(\|\hat{x}\|) < \|\hat{x}\|^2$ [16]. Let the function further satisfy

$$\psi_2(\|\hat{x}\|, \Delta t_h) + \sigma_2 \le \beta(\|\hat{x}\|) \le \psi_1(\|\hat{x}\|, \Delta t_h) \tag{34}$$

Solving (34) yields

$$\Delta t_h \ge \frac{1}{6\varrho} \min\Big\{ \ln\Big(1 + \frac{3\beta(\|\hat{x}\|)}{1 + 2\|\hat{x}\|^2}\Big), \\ -\ln\Big(1 - \frac{3(\sigma_2 + \|\hat{x}\|^2 - \beta(\|\hat{x}\|))}{1 + 5\|\hat{x}\|^2}\Big)\Big\} > 0$$

Then, the Zeno behavior is excluded. □

V. Simulation

To verify the proposed self-triggered mechanism, consider the following linear system with three players:

$$\dot{x} = Ax + B_1 u_1 + B_2 u_2 + B_3 u_3, \tag{35}$$

where $x \in \mathbb{R}^2$, $u_1, u_2, u_3 \in \mathbb{R}$, and

$$A = \begin{bmatrix} 0 & -1 \\ 2 & -3 \end{bmatrix},\ B_1 = [0; 2],\ B_2 = [0; 1],\ B_3 = [0; 3].$$

The cost functional of each player is determined by $Q_1 = 2I_{2\times 2}$, $Q_2 = I_{2\times 2}$, $Q_3 = 3I_{2\times 2}$, $R_{1j} = 2$, $R_{2j} = 1$, $R_{3j} = 3$ for $j = 1, 2, 3$. By the offline policy iteration in Algorithm 1, given the initial stabilizing feedback gains as

$$K_1^0 = [4, -2], \quad K_2^0 = [2, -1], \quad K_3^0 = [6, -3],$$

solving the cAREs in (9) yields

$$P_1^\star = \begin{bmatrix} 2.3204 & -0.3204 \\ -0.3204 & 0.3204 \end{bmatrix}, K_1^\star = \begin{bmatrix} 0.3204 & -0.3204 \end{bmatrix},$$
$$P_2^\star = \begin{bmatrix} 1.1602 & -0.1602 \\ -0.1602 & 0.1602 \end{bmatrix}, K_2^\star = \begin{bmatrix} 0.1602 & -0.1602 \end{bmatrix},$$
$$P_3^\star = \begin{bmatrix} 3.4806 & -0.4806 \\ -0.4806 & 0.4806 \end{bmatrix}, K_3^\star = \begin{bmatrix} 0.4806 & -0.4806 \end{bmatrix}.$$

The parameters in the self-triggering condition (28) are selected as $\sigma_1 = \sigma_2 = 0.5$ and $\varrho = 0.1$. With the above settings, the simulation results are shown in Figures 1 to 5.

As shown in Figure 1, the errors $\|P_i^k - P_i^\star\|$ and $\|K_i^k - K_i^\star\|$ drop to zero within 4 iterations, confirming the fast convergence of Algorithm 1 to the Nash solution to the NZS game (4). Figure 2 indicates that the closed-loop states converge quickly to the neighborhood of the origin under the intermittent implementation $u_i^\star(\hat{x}) = K_i^\star \hat{x}$ as discussed in Theorem 2, while $\hat{x}(t)$ remains piece-wise constant between two consecutive execution instants. Figure 3 shows that the three players are piecewise constant and decay as the state converges, demonstrating stable control actions under sparse updates. In Figure 4, the instantaneous costs r_i vanish with state regulation and the accumulated costs J_i converge to finite constants (0.3003, 0.1501, 0.4504), implying bounded infinite-horizon performance indices for all players. Figure 5 illustrates that the triggering intervals Δt_h stay strictly positive and tend to increase as the state approaches the equilibrium, which confirms the proposed self-triggered mechanism excludes the Zeno behavior as claimed in Theorem 2.

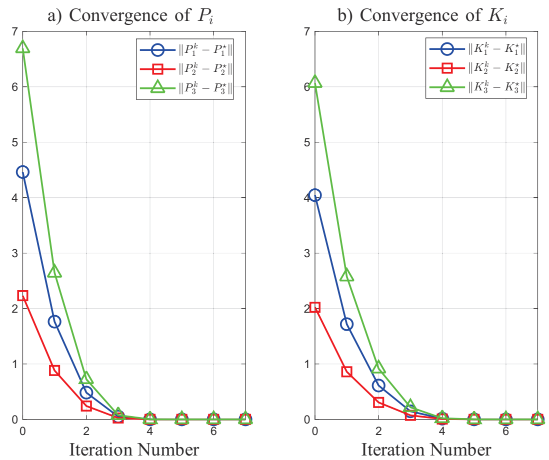

Fig. 1. Convergence of P_i and K_i.

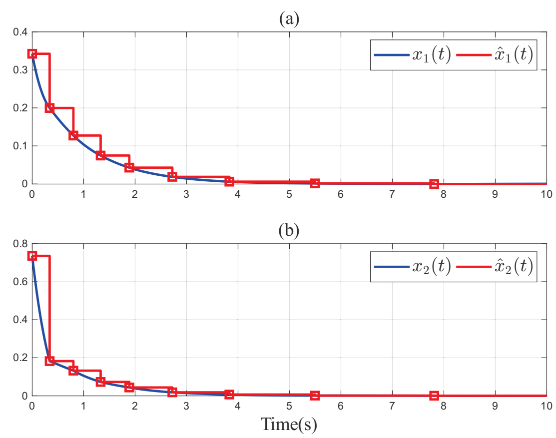

Fig. 2. Evolution of the system states and samples.

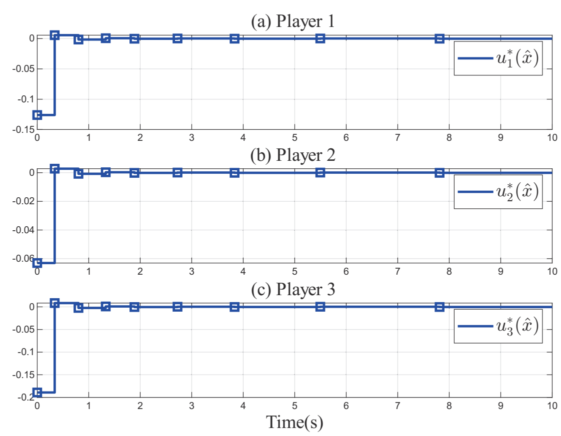

Fig. 3. Evolution of the players.

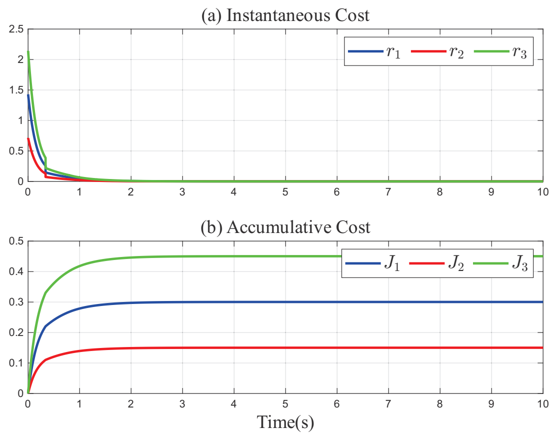

Fig. 4. Evolution of the instantaneous costs and performance indices.

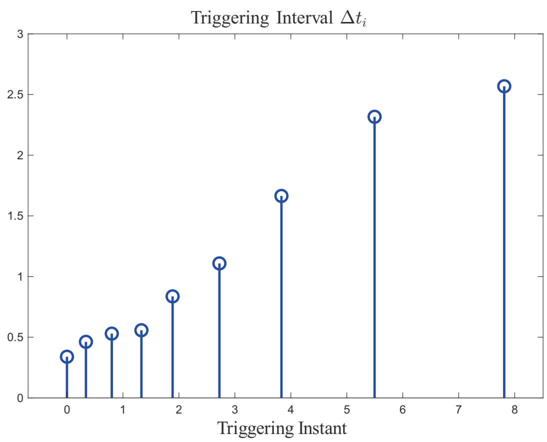

Fig. 5. Execution intervals of the proposed self-triggered mechanism.

VI. Conclusion

In this paper, a novel self-triggered mechanism is proposed for the continuous-time linear quadratic multi-player NZS game. An offline policy iteration method is implemented to solve the cAREs via alternating policy evaluation and improvement. Based on the resulting Nash feedback gains, we further develop the self-triggered implementations to eliminate the continuous state observation and thus reduce computational overhead. Rigorous theoretical analysis establishes stability guarantees together with Zeno-free execution. Simulations validate fast convergence, effective regulation, and sparse triggering with strictly positive inter-execution intervals. Future work will extend the results to model-free settings and nonlinear systems.

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